VIETNAM NATIONAL UNIVERSITY – HCMC

UNIVERSITY OF SCIENCE

Faculty of Information and Technology

Software Engineering Department

Requirement analysis

Project name: Master Interview

Course: Introduction to Software Engineering

Class: 24C10

Students:

24127186 - Giáp Đăng Khoa

24127518 - Trần Ngọc Quân

24127136 - Nguyễn Thanh Trọng

24127137 - Nguyễn Thanh Trúc

Lecturers:

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Ngô Ngọc Đăng Khoa

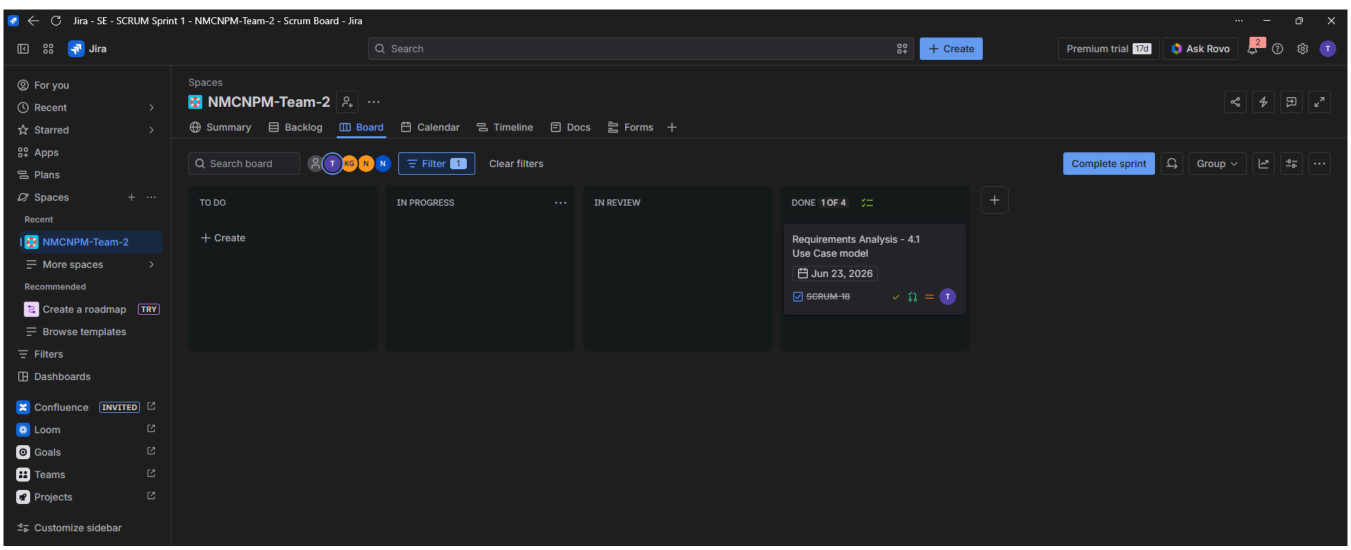
Trương Phước Lộc

Table of Contents

1 Member Contribution Assessment

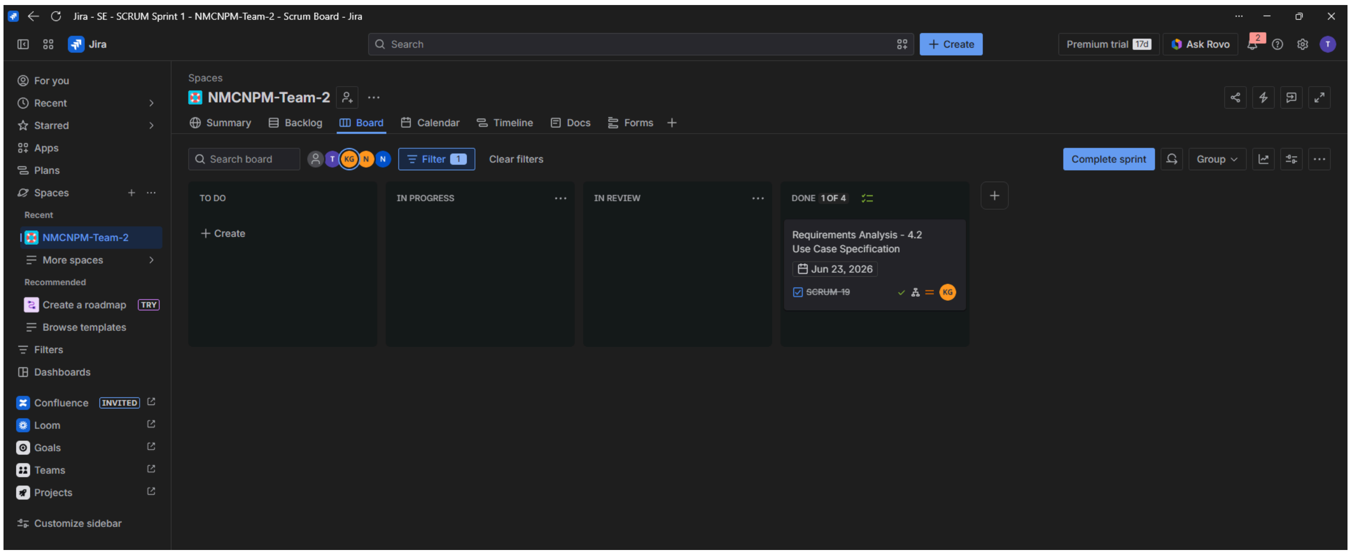
24127518 - Trần Ngọc Quân - Contribution (25%)

Task name	Summary
Design: Use Case Model	Show the core functions of system, actors, use cases, system boundaries and relations
Write Report: Use Case Model	Document the Use Case Model for the system.



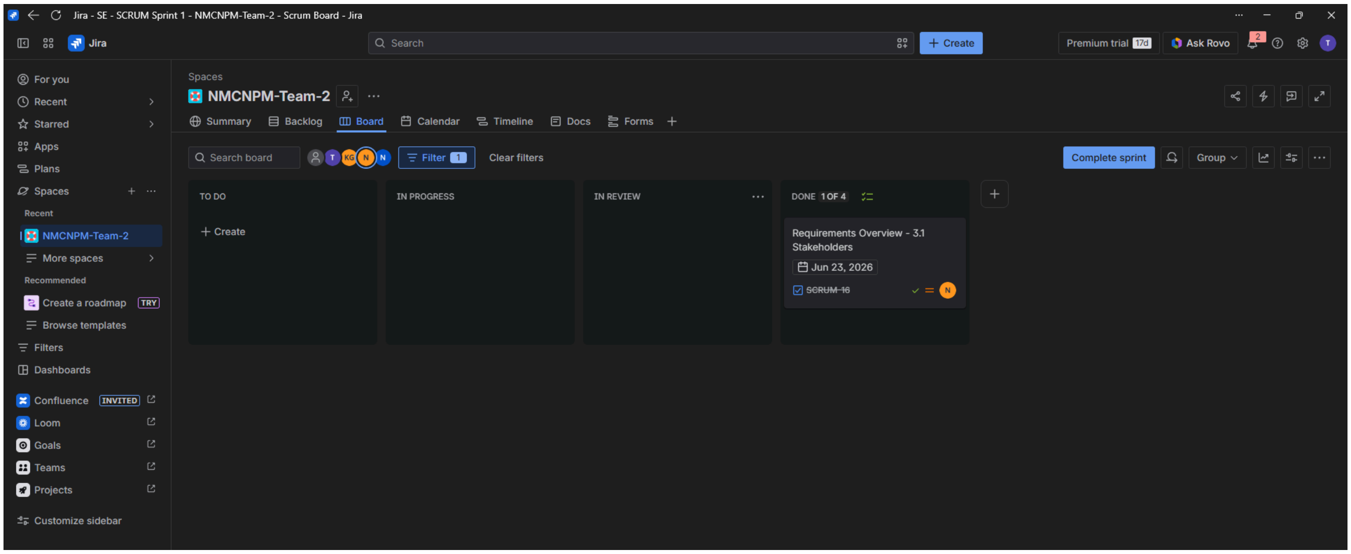
24127186 - Giáp Đăng Khoa- Contribution (25%)

Task name	Summary
Design: Use Case Specification using Figma	Create a Figma design for the Use Case Specification.
Write Report: Use Case Specification	Document the Use Case Specification



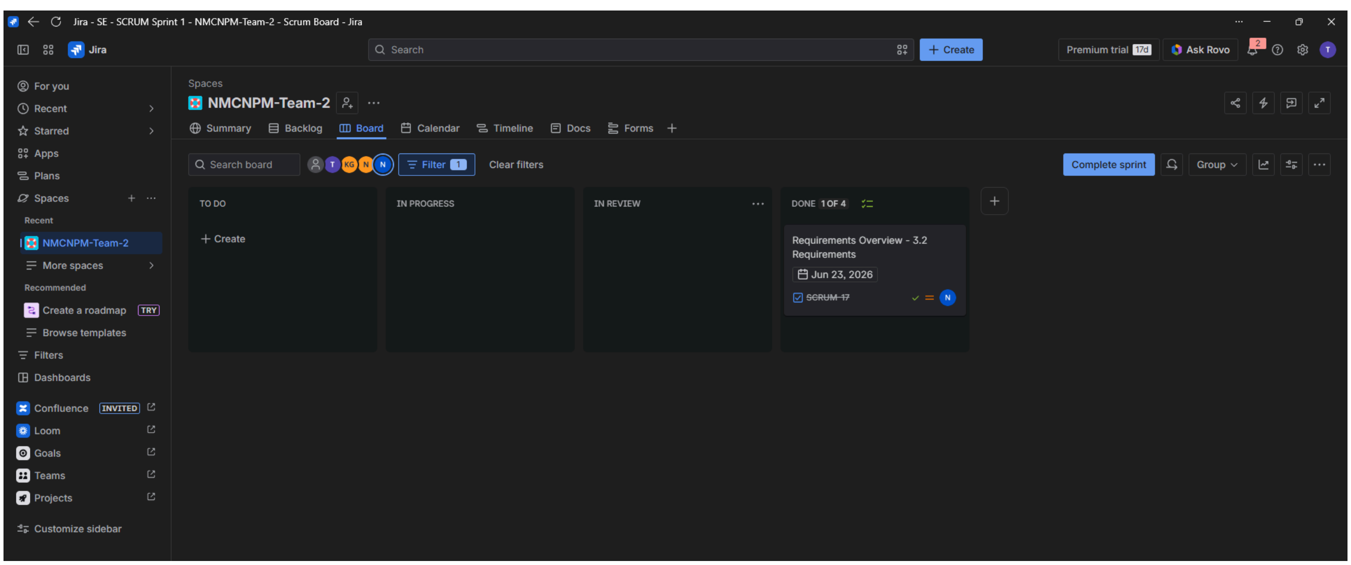
24127136 - Nguyễn Thanh Trọng - Contribution (25%)

Task name	Summary
Design: Stakeholders	Identify and document project stakeholders and their roles.
Design: Use Case Specification using Figma	Create a Figma design for the Use Case Specification.



24127137- Nguyễn Thanh Trúc - Contribution (25%)

Task name	Summary
Design: Requirements	Identify, analyze, functional and non-functional requirements for the system.
Write report: Requirements	Document Requirements



2 Problem Statement

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127186 - Giáp Đăng Khoa

Edited by:

Reviewed by:

The technology recruitment market currently faces a big gap between candidate preparation and interviewer expectations. This problem includes three main points:

- **For Interviewees (Candidates):** Recent graduates and career switchers often have good theoretical knowledge but lack practical interview skills and confidence. They do not know if their CV is good enough, what questions recruiters will ask, or how to react in a real-time conversation. Because they lack a realistic environment to practice, they fail to show their true abilities to recruiters.
- **For Interviewers (Experts):** Experienced engineers and managers want to share their knowledge and earn extra income in their free time. However, their biggest problem is **wasting time on candidates who are completely unprepared.** Without a tool to help candidates practice beforehand, experts have to spend hours on low-quality interviews, which wastes their time and makes it hard to find real talent.
- Current solutions on the market are separated and do not provide a complete preparation process. Candidates have to use different websites to fix their CV, get sample questions, or find a mentor to book a Google Meet call. There is no single platform that combines AI tools (for CV assessment, question generation, and real-time voice/text interviews) together with a direct booking system for human experts.

3 Requirements Overview

3.1 Stakeholders

Please include the ID and full name of the student who authored, reviewed, or updated this section.
Written by: 24127136 - Nguyễn Thanh Trọng
Edited by:
Reviewed by:

The student team should list (or draw a Context Diagram) and explain the role of each Stakeholder of the software.

STT	Stakeholder	Type	Description
1	Developer	Internal	The technical team or software engineer responsible for designing the system architecture, writing source code, integrating system modules, and ensuring the software effectively implements all specified functional and technical requirements.
2	Supervisor	External	The academic advisor or project supervisor. Responsible for reviewing technical quality, evaluating the software architecture, assessing the overall development process, and conducting final project acceptance.
3	Interviewee	Core Actor	The candidate (Student, Job Seeker). Uses the system to upload CVs, take assessment tests, engage in self-practice modules, and book mock interview sessions with domain experts to refine their interview skills.

4	Interviewer	Core Actor	The expert or mock interviewer. Uses the system to manage their availability slots, approve or decline booking requests, conduct live mock interviews, evaluate candidate performance, and receive payouts directly to their platform wallet.
5	Google Gemini AI	System Actor	Large Language Model (LLM) API service. Acts as the core intelligent engine to automatically parse CVs, calculate competency scores, roleplay as an AI interviewer in self-practice mode, and dynamically generate contextual questions.
6	Google Meet	System Actor	Third-party video conferencing platform integrated via API. Automatically instantiates virtual meeting rooms and dispatches the meeting invitation links to both the Interviewer and Interviewee immediately upon booking approval.
7	Stripe	System Actor	Global payment gateway. Responsible for handling secure financial transactions, including holding candidate funds upon booking request, processing immediate refunds upon cancellation, and capturing payouts to the interviewer's balance after deducting the platform commission.

3.2 Requirements

3.2.1. Functional Requirements Specification

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127137 - Nguyễn Thanh Trúc

Edited by:

Reviewed by:

ID	Module	Requirement description
F-1	**Interviewer**	
F-1.1	Authentication	The system shall allow interviewers to sign up and log in using: ● OAuth2 authentication methods. For example: Google ● Standard username and password
F-1.2	Profile setup	The system shall provide an interface for Interviewers to set up, edit, and manage their professional profiles.
F-1.3	Schedule management	The system shall allow interviewers: ● Check their current interview schedules ● Add new available time slots.
F-1.4	Mark complete interview	The system shall allow interviewers to check their current interview schedules and proactively add new available time slots.
F-1.5	Email notification	The system shall automatically send an email notification to the interviewer when a new interview is confirmed, modified, or canceled.
F-2	**Interviewee**	
F-2.1	Authentication	Interviewees shall sign up and log in using OAuth2 authentication methods. For example: Google, Github
F-2.2	Self practice	Interviewees shall be able to practice interview questions within the website questions.
F-2.3	CV evaluation	Interviewees shall send their CV and the CV is evaluated

		based on the desired job position.
F-2.4	Interview booking	Interviewee shall book an interview with a specific Interviewer at a specified time slot via the Agent.
F-2.5	Review past interview	Interviewee shall have a summary of past interviews, categorized by job positions.
F-2.6	Email confirmation	The system shall automatically send an email notification to the interviewee when an interview is successfully booked.
F-2.7	Feedback	Interviewee shall give feedback to interviewer about the attitude, professionalism
F-3	**Agent - MCP client**	
F-3.1	MCP connection	LLM agents must successfully connect with the MasterInterview website through Model Context Protocol (MCP).
F-3.2	Prompt	The agent shall handle user prompts and return desired output by working with the MCP server.
F-4	**Website - MCP server**	
F-4.1	Standard UI	The website shall have UI/UX for the users to work with.
F-4.2	MCP expose	The website backend shall expose a list of MCP tools, APIs for the agent to call.
F-4.3	Concurrency control	While the interviewee is redirected for payment, the website backend shall temporarily lock the selected time slot to prevent double-booking.
F-5	**Stripe**	
F-5.1	Sandbox payment	The system shall generate a sandbox environment to perform a money transfer demo.

3.2.2. Non-Functional Requirements Specification

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127137 - Nguyễn Thanh Trúc

Edited by:

Reviewed by:

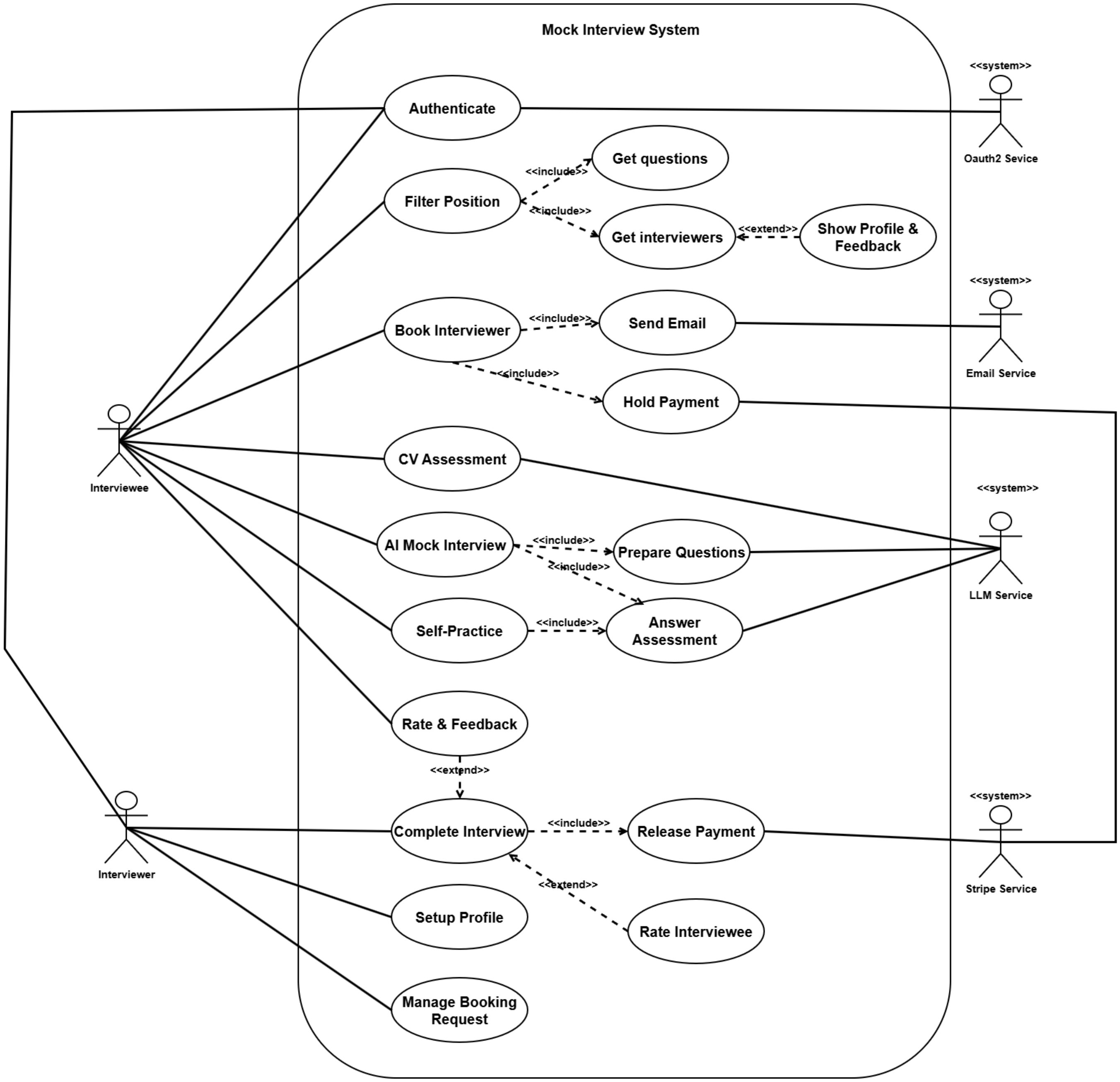
ID	Requirement name	Requirement details
NF-1	**Performance**	
NF-1.1	MCP response time	MCP server shall respond to requests from agents under 5 seconds.
NF-1.3	Webhook processing	The system shall handle signals from the Stripe sandbox and update the database within 1 second.
NF-1.4	Email latency	The server shall push the confirmed email to the mail queue within 5 seconds after successful payment.
NF-2	**Security**	
NF-2.1	MCP communication	All data between native agent and website backend via MCP shall be encoded.
NF-2.2	PII data protection	Data contains personal information (eg. CV, phone, email,etc.) shall be stored in the database or cloud storage with strict access controls.
NF-3	**Reliability**	
NF-3.1	Concurrency handling	The backend system must be thread-safe, preventing double-booking.
NF-3.2	Prevent ghost booking	The system shall temporarily lock the selected slot for **15 minutes** while the interviewee is redirected to Stripe. If the payment is not completed successfully, the system shall automatically release the lock, open the slot.

4 Requirements Analysis

4.1 Use Case model

Please include the ID and full name of the student who authored, reviewed, or updated this section.
Written by: 24127518 - Trần Ngọc Quân
Edited by:
Reviewed by:

The Use Case diagram below shows the core functions of the system. The system structure **revolves** around serving the Interviewee **through AI-driven practice features** (AI Mock Interview, Self-Practice) and **connecting them** with real Interviewers. The diagram also clearly illustrates the role of external services (OAuth2, LLM, Stripe) in **handling** the authentication flow, generating **questions**, and processing payment transactions.



4.2 *Use Case Specification*

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127186 - Giáp Đăng Khoa

Edited by:

Reviewed by: 24127136 - Nguyễn Thanh Trọng

For demonstration, please visit the link below to view the prototype in Figma

https://www.figma.com/design/c7Q8Rzja6O6vsJrC0Xagzm/Template-1?node-id=0-1&m=dev&t=eWuDNZAYSjZTaxlc-1

4.2.1. Authenticate:

Use case ID	**U001**
Use Case	Authenticate
Brief Description	The user (Interviewee or Interviewer) logs into the MasterInterview platform using an email/password or an external OAuth2 Service (like Google).
Actor	Interviewee, Interviewer, OAuth2 Service (System Actor)
Pre-Condition	1. The user has opened the MasterInterview website. 2. The user is currently logged out.
Result	The user is successfully authenticated and redirected to their specific Dashboard (Candidate or Expert Dashboard).
Main Scenario	**1. The user** clicks the "Login" button on the homepage. **2. The system** displays a login popup with two options: "Continue with Facebook" or "Continue with Google". **3. The user** clicks the "Continue with Google" button. **4. System** redirects the User to the external **OAuth2 Service** login page. **5. The user** chooses their Google account and grants permission. **6. OAuth2 Service** verifies the user's identity and sends a secure token back to the System. **7. The system** checks the token and logs the User into the platform.

	If role is interviewee: **8. The system** authenticates the User and securely passes the session token to the **Claude AI Agent**. **9. System** redirects the User to the Chat-First Dashboard. If role is interviewer: **8. System** redirects the User to Interviewer Dashboard.
Alternative Scenarios	**A1: Login Canceled by User** 1. At step 5, the User clicks "Cancel" instead of granting permission. 2. OAuth2 Service sends a cancellation message to the System. 3. System redirects the User back to the homepage and shows a message: "Login canceled. Please try again."
Non-Functional Constraints	1. **Security:** Passwords and access tokens must be highly encrypted. 2. **Performance:** The login process (from step 6 to 8) must finish in under 5 seconds.

4.2.2. Filter Position:

Use case ID	U002
Use Case	**Filter Position (Find Experts & Questions via Chat)**
Brief Description	The Interviewee types their desired job position into the AI chat. The Claude AI Agent understands the intent and displays a list of relevant interview questions and matching experts directly inside the chat interface.
Actor	Interviewee, Claude AI Agent (System Actor)
Pre-Condition	1. The Interviewee is logged into the platform.

	2. The Interviewee is on the main AI Chat interface.
Result	The chat interface successfully displays interactive UI widgets containing suitable questions and expert profiles.
Main Scenario	**1. Interviewee** types a request into the chat box (e.g., "Find me an expert for a Senior Java Developer interview" or "I want to practice Backend"). **2. Claude AI Agent** analyzes the text to extract the target job role and specific skills. **3. Claude AI Agent** calls the System's API to search the database. **4. The system** retrieves matching questions (<> Get questions) and available experts (<> Get interviewers). **5. Claude AI Agent** replies with a friendly message and displays two interactive widgets inside the chat: - A short list of practice questions. - A carousel (slider) of Expert profile cards. **6. Interviewee** clicks on an Expert's card to see more details and past ratings (<> Show Profile & Feedback).
Alternative Scenarios	**A1: Unclear Request** 1. At step 2, the AI Agent cannot detect a specific job position (e.g., the user just types "Help me find someone"). 2. Claude AI Agent replies: "Could you please specify which job role or tech stack you are looking for? (e.g., Frontend, Python, Database)." 3. The interviewee types the specific position, and the flow returns to step 2.

	A2: No Exact Matches Found 1. At step 4, the System finds no experts matching a very rare skill. 2. Claude AI Agent replies: "I couldn't find an exact match for that specific skill. However, here are some experts in related fields." 3. The system displays closely related Expert cards, and the flow continues to step 6.
Non-Functional Constraints	1. **Performance:** The AI must process the text and display the UI widgets within 5 seconds. 2. **Usability:** The Expert cards inside the chat must look clean, showing the expert's name, role, price, and a "View Profile" button.

4.2.3. Book Interviewer

Use case ID	**U003**
Use Case	Book Interviewer
Brief Description	The Interviewee requests to schedule a mock interview. The system securely holds the payment via Stripe and sets the booking to "Pending". The AI Agent notifies the Interviewee once the Expert approves or declines the request.
Actor	Interviewee, Stripe Service (System Actor), Email Service (System Actor)
Pre-Condition	1. The Interviewee is logged into the platform.

	2. The Interviewee is viewing an Expert's profile or a booking widget inside the AI chat.
Result	A booking is confirmed, funds are held, and both parties receive a Google Meet link for the interview session.
Main Scenario	**1. Interviewee** selects a suitable date and time from the Expert's calendar widget, then clicks "Confirm & Pay". **2. System** calculates the total fee and opens a secure payment form powered by **Stripe Service**. **3. Interviewee** enters their credit card details and submits the payment. **4. Stripe Service** verifies the card and securely places a temporary hold (Auth) on the funds. **5. System** saves the new booking record into the database with a **"Pending Approval"** status. **6. System (AI Agent)** updates the chat interface: *"Your booking request has been sent! I will let you know as soon as the Expert confirms."* **7. System** detects that the Expert has accepted the booking request. **8. System** updates the booking status to "Scheduled" and generates a unique Google Meet link. **9. Email Service** automatically sends a confirmation email (containing the Google Meet link and calendar invite) to both the Interviewee and the Expert (<> Send Email). **10. System (AI Agent)** sends a push notification in the chat (or greets the user on their next login): *"Great news! Your interview with [Expert Name] is confirmed. Check your email for the Meet link!"*
Alternative Scenarios	**A1: Payment Failure (Stripe Rejection)** 1. At step 6, the Stripe Service declines the card (e.g., insufficient funds or wrong CVV).

	2. System stops the booking process and displays an error message: "Payment failed. Please check your card details and try again." 3. Interviewee enters a different card, and the flow returns to step 5. **A2: Time Slot Conflict** 1. At step 3, the System checks the database and finds that another user has just booked that exact time slot. 2. System displays a warning: "This time slot is no longer available. Please choose another time." 3. Interviewee selects a new time slot from the calendar, and the flow continues. **A3. Expert Declines or Request Expires (7 Days)** 1. At step 7, the Expert explicitly declines the request, or 7 days pass without any action. 2. System updates the status to "Canceled". 3. System calls Stripe Service to release the payment hold immediately (no charges applied). 4. **Email Service** sends a cancellation email to the Interviewee, confirming that the request was declined and no funds were charged. 5. System (AI Agent) notifies the Interviewee: *"Unfortunately, the Expert couldn't accept your request at this time. Your held funds have been fully released."*
Non-Functional Constraints	1. **Security:** All payment transactions must strictly follow PCI-DSS security standards. The System must not save the actual credit card numbers in its database. 2. **Reliability:** If the Email Service is temporarily down at step 8, the System must still save the booking and automatically retry sending the email later.

4.2.4. Curriculum Vitae Assessment:

Use case ID	U004
Use Case	CV Assessment
Brief Description	The Interviewee uploads their resume (CV) into the system. The LLM Service analyzes the document and provides a detailed feedback report, including a matching score for a specific job role.
Actor	Interviewee, LLM Service (System Actor).
Pre-Condition	1. The Interviewee is logged into the platform. 2. The Interviewee has a valid CV file (PDF or DOCX format).
Result	The CV is successfully parsed, evaluated, and a structured feedback report is displayed to the Interviewee.
Main Scenario	**1. Interviewee** drags and drops their CV file into the AI chat interface or clicks the upload button. **2. System** validates the file format and extracts the text content from the document. **3. System** sends the extracted CV text along with the target job position to the **LLM Service** via API. **4. LLM Service** analyzes the data to identify the candidate's strengths, weaknesses, and missing skills compared to the job requirements. **5. LLM Service** returns a structured JSON response containing the assessment results and a matching score (e.g., 85/100) to the System.

	6. The system renders the data into an interactive UI widget inside the chat. **7. Interviewee** views the detailed feedback and actionable suggestions for improvement.
Alternative Scenarios	**A1: Unsupported File Format or File Too Large** 1. At step 2, the System detects that the file is an image (.jpg) or exceeds the 5MB size limit. 2. System rejects the file and displays an error message: "Invalid file format or size. Please upload a PDF or DOCX under 5MB." 3. Interviewee uploads a correct file, and the flow restarts. **A2: LLM Service Timeout** 1. At step 4, the LLM Service takes too long to respond or is temporarily down. 2. The system displays a warning: "The AI analysis is taking longer than expected. Please try again later." 3. The process is aborted.
Non-Functional Constraints	1. **Performance:** The text extraction and AI analysis process must be completed within 10 seconds. 2. **Privacy:** The System must explicitly state that the uploaded CV data will not be used to train external public AI models.

4.2.5. *AI Mock Interview*

Use case ID	U005
Use Case	AI Mock Interview
Brief Description	The Interviewee participates in a simulated, interactive interview session with the AI Agent. The system dynamically generates questions and provides real-time evaluation of the candidate's answers.
Actor	Interviewee, LLM Service (System Actor)
Pre-Condition	1. The Interviewee is logged in and is on the AI Chat interface. 2. The Interviewee has selected a specific job role to practice.
Result	A complete mock interview session is conducted, and the Interviewee receives a final performance assessment.
Main Scenario	**1. Interviewee** clicks the "Start AI Mock Interview" button inside the chat. **2. System** sends the selected job profile (and the user's CV if available) to the **LLM Service** to generate context-aware questions (<> Prepare Questions). **3. LLM Service** acts as the Interviewer, sending the first technical question via the chat interface. **4. Interviewee** types or records their answer and sends it. **5. System** forwards the candidate's response to the **LLM Service** for real-time grading (<> Answer Assessment). **6. LLM Service** analyzes the answer, provides brief constructive feedback, and immediately asks a follow-up question. **7. System** saves the interaction into the database.

	8. Steps 4 to 7 are repeated until the predefined number of questions is reached. **9. System** concludes the session and displays a final comprehensive report widget.
Alternative Scenarios	**A1: User Requests a Hint or Skips** 1. At step 4, the Interviewee types "I don't know" or "Can you give me a hint?". 2. LLM Service provides a small conceptual hint or explicitly explains the correct answer. 3. System records a lower score for that specific question. 4. LLM Service moves on to the next question. **A2: Inappropriate Content Detected** 1. At step 4, the Interviewee sends a message containing profanity or off-topic content. 2. System filters the message before sending it to the LLM Service. 3. System issues a warning to the Interviewee to maintain professional conduct during the mock interview.
Non-Functional Constraints	1. **Latency:** The LLM Service must generate and return the next question or assessment within 3-4 seconds to maintain a natural conversation flow. 2. **Context Retention:** The AI must remember the context of previous answers throughout the entire interview session.

4.2.6. Self-Practice:

Use case ID	**U006**
Use Case	Self-Practice with AI chat-bot.
Brief Description	The Interviewee selects a specific interview question from the platform's question bank and submits their answer. The LLM Service evaluates the response and provides immediate, targeted feedback.
Actor	Interviewee, LLM Service (System Actor).
Pre-Condition	1. The Interviewee is logged into the system. 2. The Interviewee has found a specific question they want to practice (e.g., via the "Filter Position" feature).
Result	The Interviewee's specific answer is analyzed, graded, and a feedback report is displayed.
Main Scenario	**1. Interviewee** types a request into the chat box to practice a specific topic or question (e.g., "I want to self-practice a question about SQL Joins"). **2. System (Claude AI Agent)** analyzes the natural language input, detects the intent, and retrieves a relevant question from the database. **3. The system** opens an interactive practice widget inside the chat, displaying the chosen question. **4. The interviewee** types out their answer (or records their voice) and clicks "Submit Answer". **5. The system** forwards both the question and the candidate's answer to the **LLM Service** for evaluation (<> Answer Assessment). **6. LLM Service** analyzes the answer for technical accuracy, clarity, and completeness.

	7. LLM Service returns a detailed assessment, including a score out of 10, areas for improvement, and a sample "Model Answer". **8. System** renders the feedback data and displays it to the Interviewee.
Alternative Scenarios	**A1: Missing Topic or Difficulty Context** 1. At step 2, the AI Agent detects the intent but is missing parameters (e.g., user types "Give me a hard question" or "I want to practice SQL"). 2. System (AI Agent) prompts the user for the missing information: "Sure! What specific topic and difficulty level (Easy, Medium, Hard) would you like to practice?" 3. Interviewee provides the missing details, and the flow seamlessly continues to step 2. **A2: Incomplete or Empty Answer** 1. At step 3, the Interviewee submits a blank text box or an extremely short answer (e.g., "I don't know"). 2. System validates the input length before sending it to the LLM. 3. The system displays a prompt: "Your answer is too short. Please try to explain your thoughts in more detail to get an accurate assessment." 4. The interviewee modifies their answer and submits again. **A3: Voice Recording Failure** 1. At step 3, the Interviewee tries to use the microphone, but browser permission is denied. 2. System shows an alert: "Microphone access blocked. Please enable it in your browser settings or switch to text input." **A3: User Requests Model Answer Directly (Skip Answering)**

	1. At step 3, instead of typing an answer, the Interviewee clicks the "Show Model Answer" button (or types "Show me the answer"). 2. The system skips the "Answer Assessment" step since there is no user input to grade. 3. The system calls the LLM Service to generate (or retrieves from the database) the ideal answer for that specific question. 4. The system displays the detailed Model Answer to the Interviewee without any grading score.
Non-Functional Constraints	1. **Usability:** The input area must support code formatting (Markdown) so candidates can type code snippets for technical questions. 2. **Performance:** The feedback generation must be completed within 3 seconds to maintain a smooth learning flow.

4.2.7. Rate & Feedback for Expert:

Use case ID	U007
Use Case	Rate & Feedback
Brief Description	After finishing a mock interview session with a human Expert, the Interviewee provides a star rating and a written review based on their experience.
Actor	Interviewee
Pre-Condition	1. The Interviewee is logged into the platform. 2. A scheduled mock interview session with an Expert has just been completed.
Result	The feedback is saved, and the Expert's overall rating on their profile is updated.

Main Scenario	**1. System** detects that the scheduled interview session has officially ended. **2. System** displays a feedback popup or widget inside the user's dashboard. **3. Interviewee** selects a star rating (from 1 to 5 stars) indicating their satisfaction level. **4. Interviewee** types a written review (optional) detailing the Expert's performance and helpfulness. **5. Interviewee** clicks the "Submit Feedback" button. **6. System** saves the rating and review data into the database. **7. System** recalculates the Expert's average rating score. **8. System** displays a "Thank You" message and returns the Interviewee to the main homepage.
Alternative Scenarios	**A1: Interviewee Skips Feedback** 1. At step 3, the Interviewee clicks the "Skip for now" button instead of rating. 2. System closes the popup without saving any data. 3. System places a small notification badge on the Dashboard to remind the Interviewee to leave feedback later. **A2: System Submission Error** 1. At step 5, a database or network error occurs when the user clicks submit. 2. System displays an error message: "We couldn't save your feedback due to a connection issue. Please try again."

	3. Interviewee checks their connection and clicks "Submit" again.
Non-Functional Constraints	1. **Data Integrity:** An Interviewee is strictly allowed to review an Expert *only* if they have successfully completed a paid booking with that specific Expert (to prevent fake reviews). 2. **Real-time Update:** The Expert's average star rating must update immediately on the "Filter Position" page once new feedback is submitted.

4.2.8. Manage Booking Requests:

Use case ID	U008
Use Case	Manage Booking Requests
Brief Description	The Interviewer reviews pending mock interview requests from candidates and chooses to either accept or decline them.
Actor	Interviewer, Email Service (System Actor)
Pre-Condition	1. The Interviewer is logged into the platform. 2. There is at least one "Pending" booking request in their dashboard.
Result	The booking status is updated to either "Scheduled" or "Canceled", and the appropriate follow-up actions (Email/Stripe) are triggered.
Main Scenario	**1. The interviewer** navigates to the "Booking Requests" section on their dashboard. **2. The system** displays a list of pending requests, showing the candidate's name, requested time, and included message/CV. **3. The interviewer** clicks on a specific request to view details.

	4. The interviewer clicks the "Accept Request" button. **5. System** updates the booking status in the database to "Scheduled". **6. The system** generates a unique Google Meet link for the session. **7. Email Service** sends a confirmation email with the calendar invite to both the Interviewer and the candidate. **8. The system** removes the request from the "Pending" list and moves it to the "Upcoming Interviews" list.
Alternative Scenarios	**A1: Interviewer Declines Request** 1. At step 4, the Interviewer clicks the "Decline Request" button. 2. The system prompts the Interviewer to select an optional reason (e.g., "Schedule Conflict"). 3. The system updates the booking status to "Canceled". 4. The system triggers the Stripe Service to release the candidate's held funds immediately. 5. Email Service sends a cancellation email to the candidate.
Non-Functional Constraints	1. **Time Limit:** If the Interviewer takes no action within 7 days, the system must automatically execute the Decline flow (A1) to comply with Stripe's authorization hold limits. 2. **Real-time Sync:** The dashboard must reflect status changes immediately without requiring a page reload.

4.2.9. Manage Availability:

Use case ID	**U009**

Use Case	Manage Availability
Brief Description	The Interviewer sets up and manages their available time slots for mock interviews. The system updates their public calendar so candidates can book available sessions.
Actor	Interviewer
Pre-Condition	1. The Interviewer is logged into the platform. 2. The Interviewer's profile is approved and active.
Result	The Interviewer's availability is updated in the database and reflected on their public booking page.
Main Scenario	**1. The interviewer** navigates to the "Calendar Settings" section on their dashboard. **2. The system** displays a weekly calendar interface with current active time slots. **3. The interviewer** selects specific days of the week and adds available time blocks (e.g., Monday 19:00 - 21:00). **4. The interviewer** (Optional) applies rules such as "Repeat weekly" or sets buffer times between sessions. **5. The interviewer** clicks the "Save Schedule" button. **6. The system** validates the selected time slots to ensure there are no formatting errors. **7. The system** saves the new availability into the database. **8. The system** displays a success message and immediately updates the Interviewer's public booking widget.
Alternative Scenarios	**A1: Time Slot Conflict with Existing Bookings**

	1. At step 6, the System detects that the Interviewer has deleted a time slot or changed their availability in a way that conflicts with an already "Scheduled" interview. 2. The system displays a warning: "You have an existing booking during this time. Please cancel or reschedule the booking before removing this time slot." 3. The conflicting change is not saved, and the Interviewer must manually resolve it. **A2: Invalid Time Selection** 1. At step 6, the System detects that the End Time is earlier than the Start Time (e.g., 20:00 to 18:00). 2. The system displays an inline error message: "End time must be after start time." 3. The interviewer corrects the input and clicks "Save" again.
Non-Functional Constraints	1. **Timezone Conversion:** The system must automatically store the Interviewer's availability in UTC standard within the database, and flawlessly convert it to the Candidate's local timezone when displayed on the public profile. 2. **Usability:** The calendar UI should support drag-and-drop actions for intuitive time selection.

4.2.10. Complete Interview:

Use case ID	U010
Use Case	Complete interview

Brief Description	After conducting the mock interview, the Interviewer marks the session as complete. The system then captures the previously held funds via Stripe, deducts the platform fee, and credits the Interviewer's balance.
Actor	Interviewer, Stripe Service (System Actor)
Pre-Condition	1. The interview session is in the "Scheduled" status. 2. The scheduled time for the interview has passed or the session has just concluded.
Result	The booking status is changed to "Completed", and the Interviewer's wallet is updated.
Main Scenario	**1. The interviewer** navigates to their dashboard after finishing the Google Meet session. **2. The system** displays the current/recent booking and provides an action button. **3. The interviewer** clicks the "Mark as Completed" button for that specific session. **4. System** updates the booking status in the database to "Completed". **5. System** triggers the **Stripe Service** to execute the "Capture" action on the Candidate's temporarily held funds. **6. Stripe Service** processes the transaction, deducts the platform's commission fee, and credits the remaining amount to the Interviewer's payout balance. **7. The system** displays a success message showing the total earnings from the session. **8. The system** automatically triggers the Post-Interview flow (Use Case 7) to send emails and request feedback.

Alternative Scenarios	**A1: Candidate No-Show** 1. At step 3, the Interviewer clicks the "Candidate No-Show" button instead. 2. The system updates the status to "No-Show". 3. The system still calls Stripe to capture the funds to compensate the Interviewer's wasted time. 4. Flow continues to step 6.
Non-Functional Constraints	1. Audit Trail: Every financial transaction (Capture, Fee Deduction, Payout) must be securely logged in the database for accounting and dispute resolution.

4.2.11. Setup profile for interviewer:

Use case ID	**U011**
Use Case	Setup & Manage Profile
Brief Description	The Interviewer sets up or updates their public profile, defines their interview rate, and uploads verification documents (PDFs) for Admin approval. The profile dashboard also provides a financial overview of their earnings.
Actor	Interviewer, Cloud Storage Service (System Actor), Admin.
Pre-Condition	The Interviewer is successfully logged into the platform.
Result	Profile data is saved, media/documents are securely uploaded to cloud storage, and the profile status is updated to "Pending Review" (or published directly if applicable).
Main Scenario	1. **Interviewer** navigates to the "Profile & Dashboard" page.

	2. **System** displays the financial overview widget (Total Earnings, Pending Balance) and the profile configuration form. 3. **Interviewer** inputs personal details (Name, Title, Bio) and uploads a profile picture (Avatar). 4. **Interviewer** sets their desired rate in the "Price per Interview" field. 5. **System** dynamically calculates and displays the net earnings after deducting the platform fee (e.g., "You will earn $24 after the 20% platform fee"). 6. **Interviewer** uploads verification documents (CV, Degrees, IT Certifications) in PDF format into the "Trust Center" section. 7. **Interviewer** clicks the "Submit for Review" button. 8. **System** calls the **Cloud Storage Service** to securely upload the image and PDF files, retrieving their respective URLs. 9. **System** saves the text data, pricing, and file URLs into the Database. 10. **System** updates the profile status to "Pending Review", triggers a notification to the **Platform Admin** to verify the PDF documents, and displays a success message to the Interviewer.
Alternative Scenarios	**A1: Invalid File Size or Format** 1. At step 6 or 8, the System detects that the uploaded Avatar/PDF exceeds the size limit or has an invalid file extension. 2. System halts the upload process and displays an inline error (e.g., "PDF must be under 10MB"). 3. Interviewer selects a valid file and retries. **A2: Minor Update (No Review Required)**

	1. At step 10, if the Interviewer is already verified and only updates basic info (e.g., Bio, Avatar) WITHOUT changing the interview rate or uploading new PDFs. 2. System skips the "Pending Review" phase. 3. System immediately updates the active public profile and displays: "Your profile has been successfully updated."
Non-Functional Constraints	1. **Data Privacy:** Verification PDFs contain sensitive personal data. They must only be accessible by the Admin and the owner, and just display publicly to Candidates by badges. 2. **Data Integrity:** The "Price per Interview" field must be strictly validated as a positive integer (>0) to prevent logic errors during Stripe payment processing. 3. **Performance & Architecture:** All media files (images, PDFs) must be handled by the Cloud Storage Service. The Database must only store textual data and file URLs to ensure optimal query performance.

5 AI Usage Declaration

Based on the provided AI Usage Guideline document, teams must submit their AI usage declaration here. Students are encouraged to leverage AI tools creatively and effectively in this project.

Gemini. Gemini, Google, gemini.google.com, accessed [14:39 on June 18, 2026], prompt: "Given functional requirements: Interviewer: read interviewee profile... Describe the functional requirements of the system using natural language.", used for the [functional requirement analysis]; AI provided the structured functional requirements categorized by system modules, students translated the output into standard software engineering table and incorporated it into their project specifications.

Gemini. Gemini, Google, gemini.google.com, accessed [21:50 on June 20, 2026], prompt: "Non-functional requirements for this system, below is my current ideas: System respond time to agent, Time since success booking to send confirmation email, Latency for money transaction", used for the [non-functional requirement analysis]; AI provided the structured functional requirements categorized by performance, security, reliability, students selected the suitable output to non functional requirement table

Gemini. Gemini, Google, gemini.google.com, accessed [08:45 on June 18, 2026], prompt: "Use Case Diagram of the system là gì và dùng để làm gì", used for the [use case diagram construction]; AI provided clear explanations of actors, use cases, and system boundaries, students applied these guidelines to correctly identify the entities and design the formal Use Case Diagram for the mock interview system. .

6 Presentation

Teams are required to record presentation videos of their work using the provided template.

Every team member must participate in the presentation, specifically covering the sections their contributed to the project.

Videos must not exceed 30 minutes.

Please upload your videos to Youtube (set to either Unlisted or Public).

Provide the Youtube links the the field below.

https://youtu.be/8uZSvIXDHtE

7 Reflective Report

Most Helpful for Implementation: Sections 3.2 (Requirements Overview) and 4 (Requirements Analysis) are highly practical as they serve as the functional blueprint for the project.

- **Reasoning:** These sections explicitly define what needs to be built and outline the core features of the system. For instance, having a clear requirements analysis allows the development team to understand the exact scope of work, research the necessary technologies beforehand, and proceed directly with implementation knowing exactly what functionalities to code.

Unnecessary / Less Practical: Section 3.1 (Stakeholder) is mostly theoretical and lacks practical value for the direct implementation of our project.

- **Reasoning:** Identifying stakeholders does not have a direct, functional impact on the software development process itself. Since we are a team building an academic project rather than a commercial product for a complex network of clients, analyzing stakeholders does not contribute to the actual coding, system design, or feature optimization.